2-µm 5-kHz PM DFB-FBG fiber seed laser in ns pulsed Tm-doped fiber amplifier

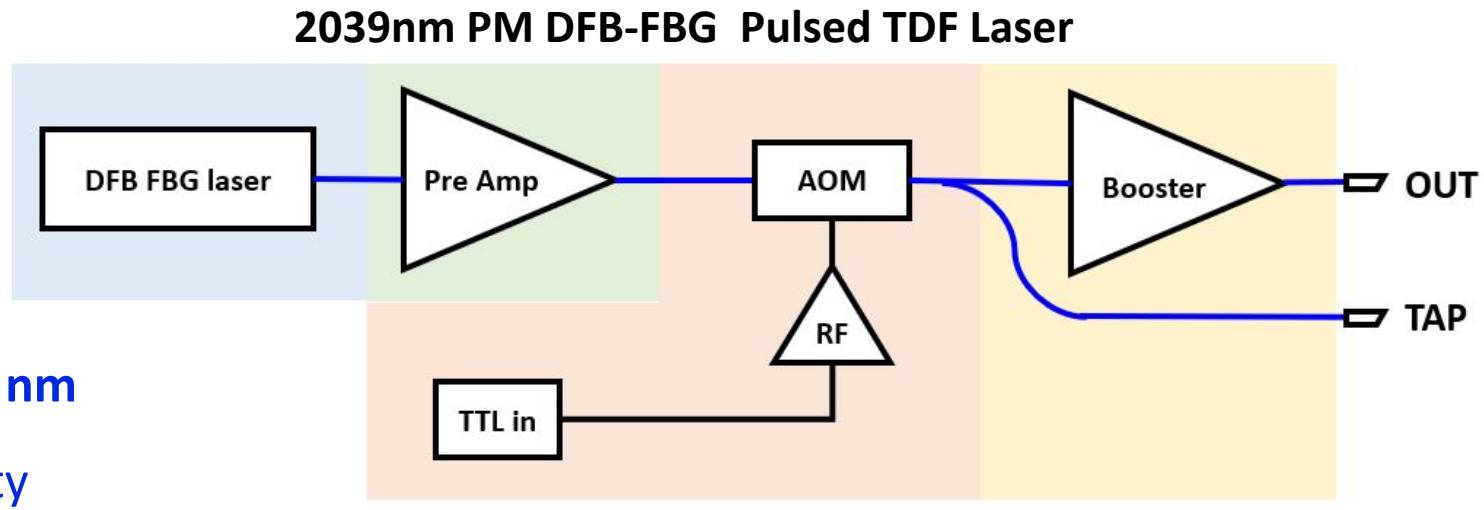
Wiktor Walasik[(1)], **Robert E. Tench**[(1)], **Alexandre Amavigan**[(1)], **Jean-Marc Delavaux**[(1)], **and Andre Van Rynbach**[(2)]

wiktor.walasik@cybel-llc.com

[(1)]CYBEL LLC, Bethlehem, PA, USA

[(2)]Air Force Research Laboratory, Dayton, OH

Talk Outline



- **Motivation and Objectives**
- **DFB-FBG laser performance at 2051 nm**

  Output power, spectrum, stability
- **5-kHz 2039 nm PM DFB-FBG as a CW seed in TDF preamplifier**

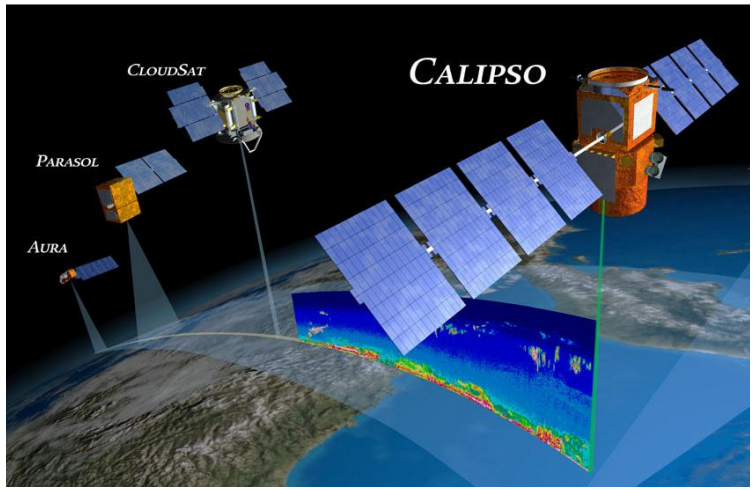
  Output power, OSNR, NF, stability & linewidth
- **50-ns pulse performance of 50 W pulsed PM TDF booster**

  Peak power, pulse energy, spectrum, stability
- **Conclusions**

Motivation and Objectives

DFB-FBG fiber laser features:

- Large available wavelength choice in 2 µm range
- High power possible (> 60 mW)
- Narrow linewidth (5 kHz)
- Single frequency operation
- Good wavelength stability (20 pm/^{o}C)
- Compact, all-fiber design
- SM or PM design



Applications:

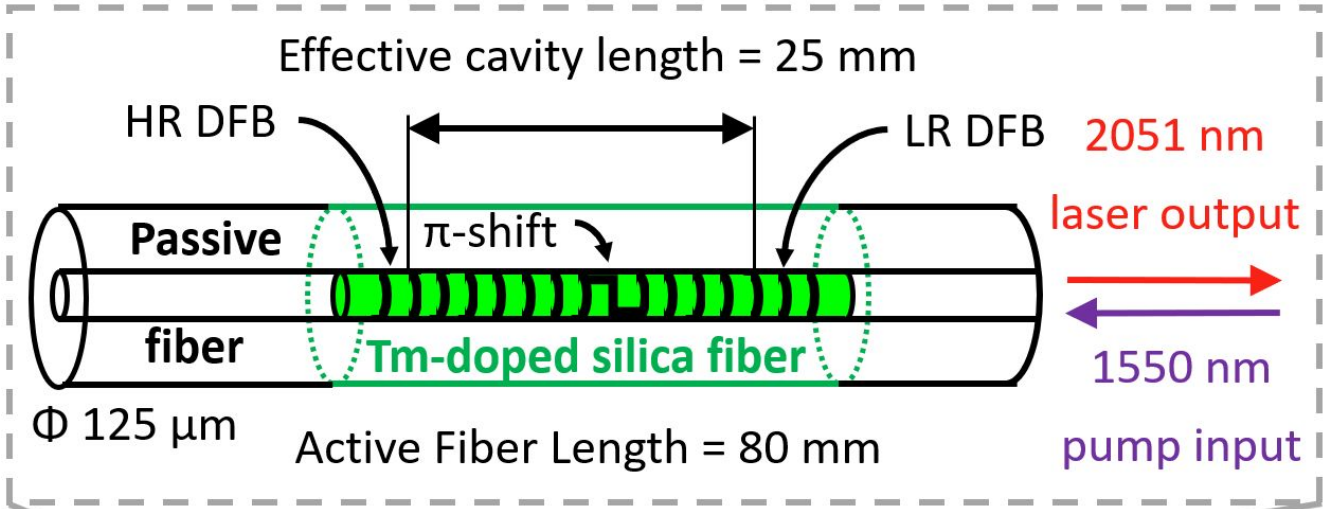
- Light Detection and Ranging (LIDAR)
- Optical sensors
- Spectroscopy
- Telecommunications
- Medicine

Source: NASA

2 µm DFB-FBG vs Semiconductor seed laser comparison

Parameter	DFB-FBG laser	Semiconductor laser
Fabrication	**easy print process**	multistep process
Output power	**tens of mW**	few mW
Wavelength tuning with power [pm/mW]	**≈0.01 pm/mW**	≈10 pm/mW
Temp. tuning [pm/°C] [GHz/°C]	≈20 ≈1.4	≈100 ≈7.1
Linewidth	**tens of kHz**	**tens of kHz to few MHz**
Modulation scheme	external	direct or external

DFB-FBG concept and operation



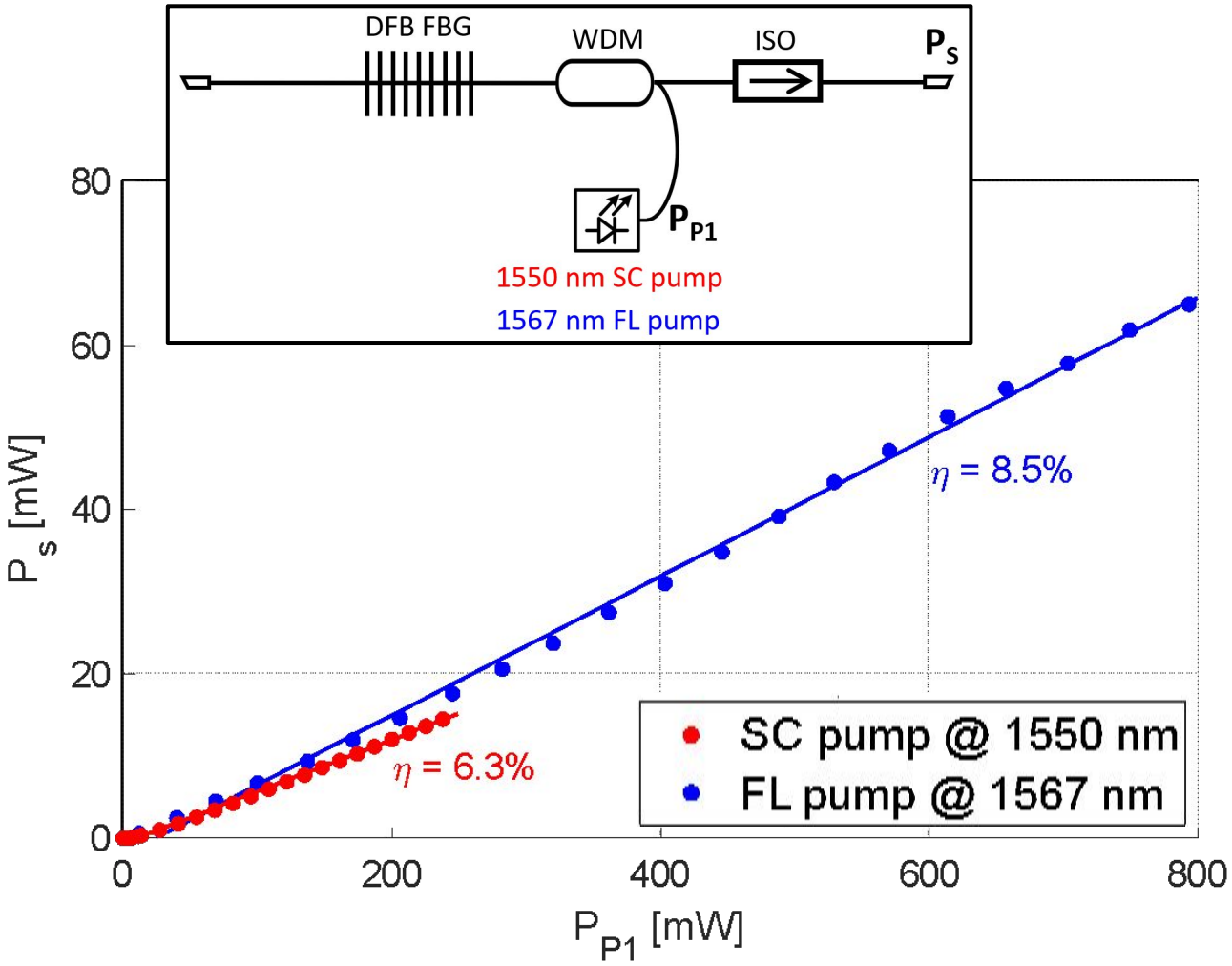
DFB FBG concept:

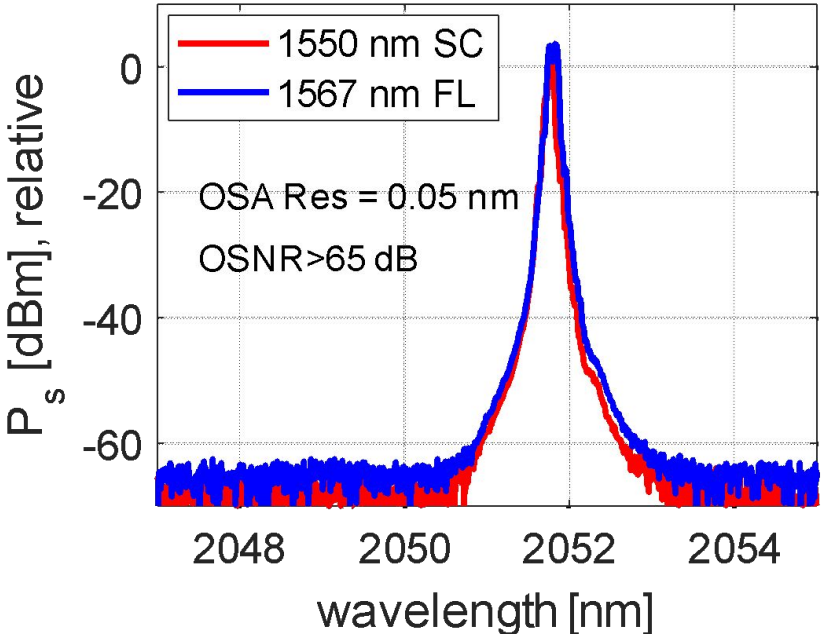
- Single 60-mm-long apodized Distributed Feed-Back Fiber Bragg Grating
- π phase shift in the center
- 4 GHz longitudinal mode spacing

Pumping configuration:

- Counter-pumping
- 250 mW Semiconductor laser (SC) @ 1550 nm, or
- 1W Fiber Laser (FL) @ 1567 nm

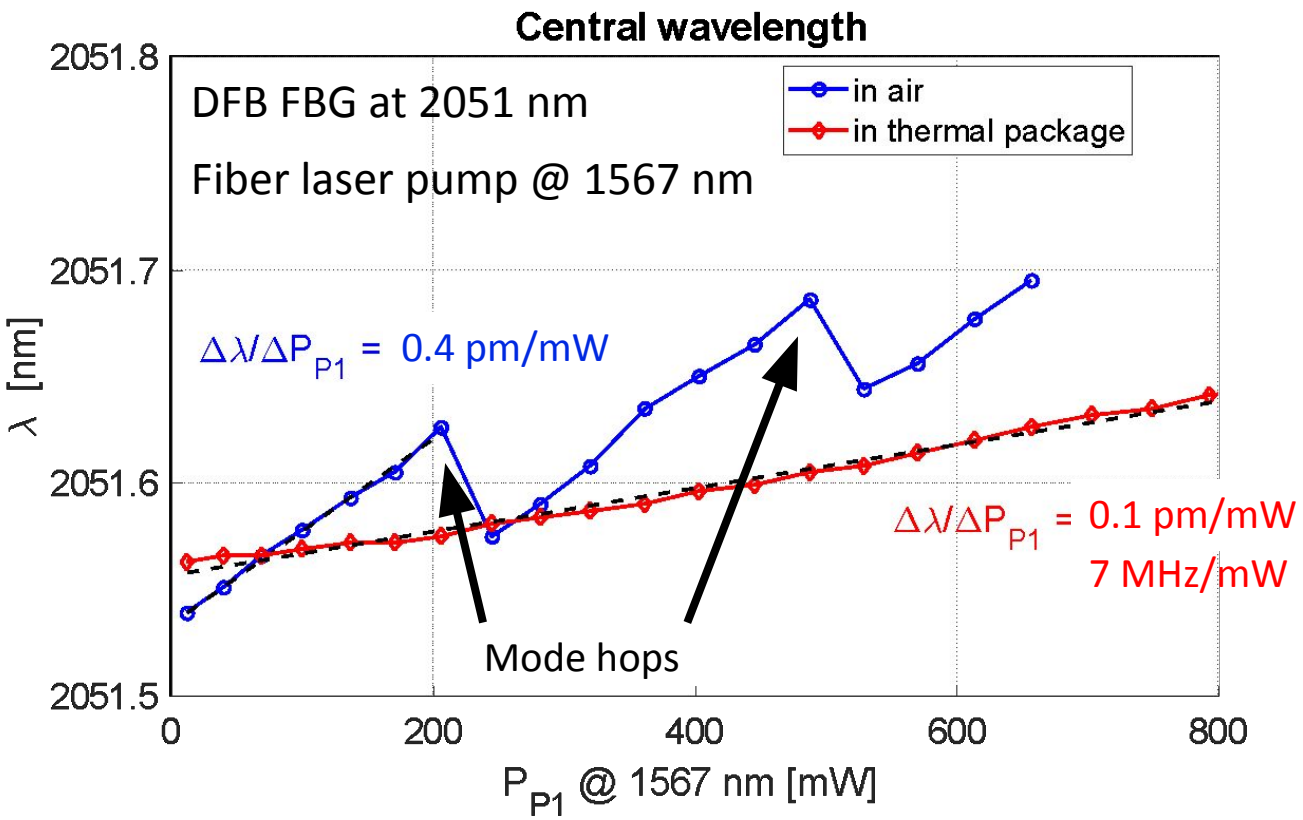
LL curves for SC and FL pumping

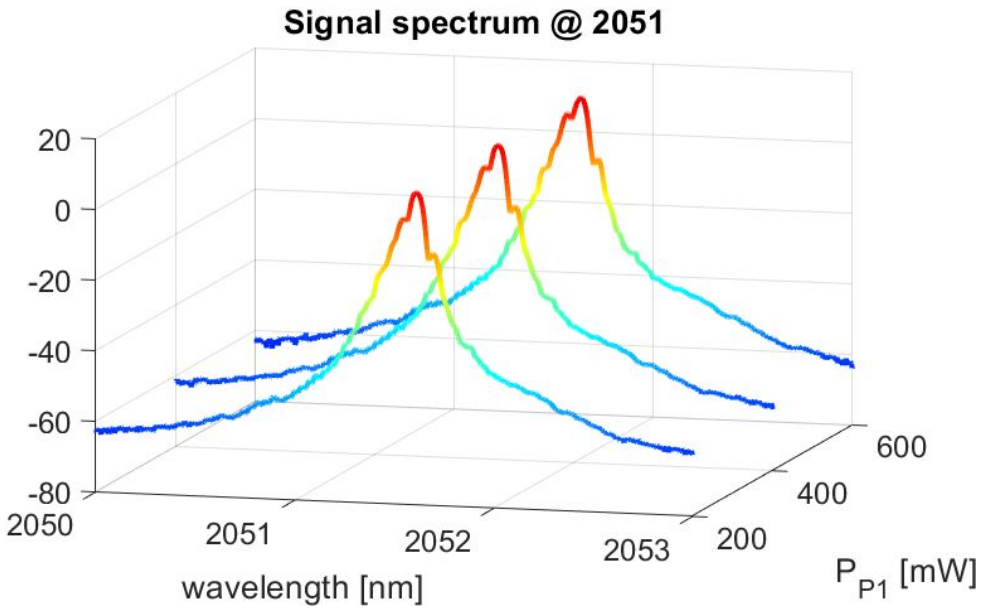




- **Single mode operation for up to 65 mW**
- **Up to 10% slope efficiency**
- **No sidebands**

DFB-FBG wavelength vs pump power

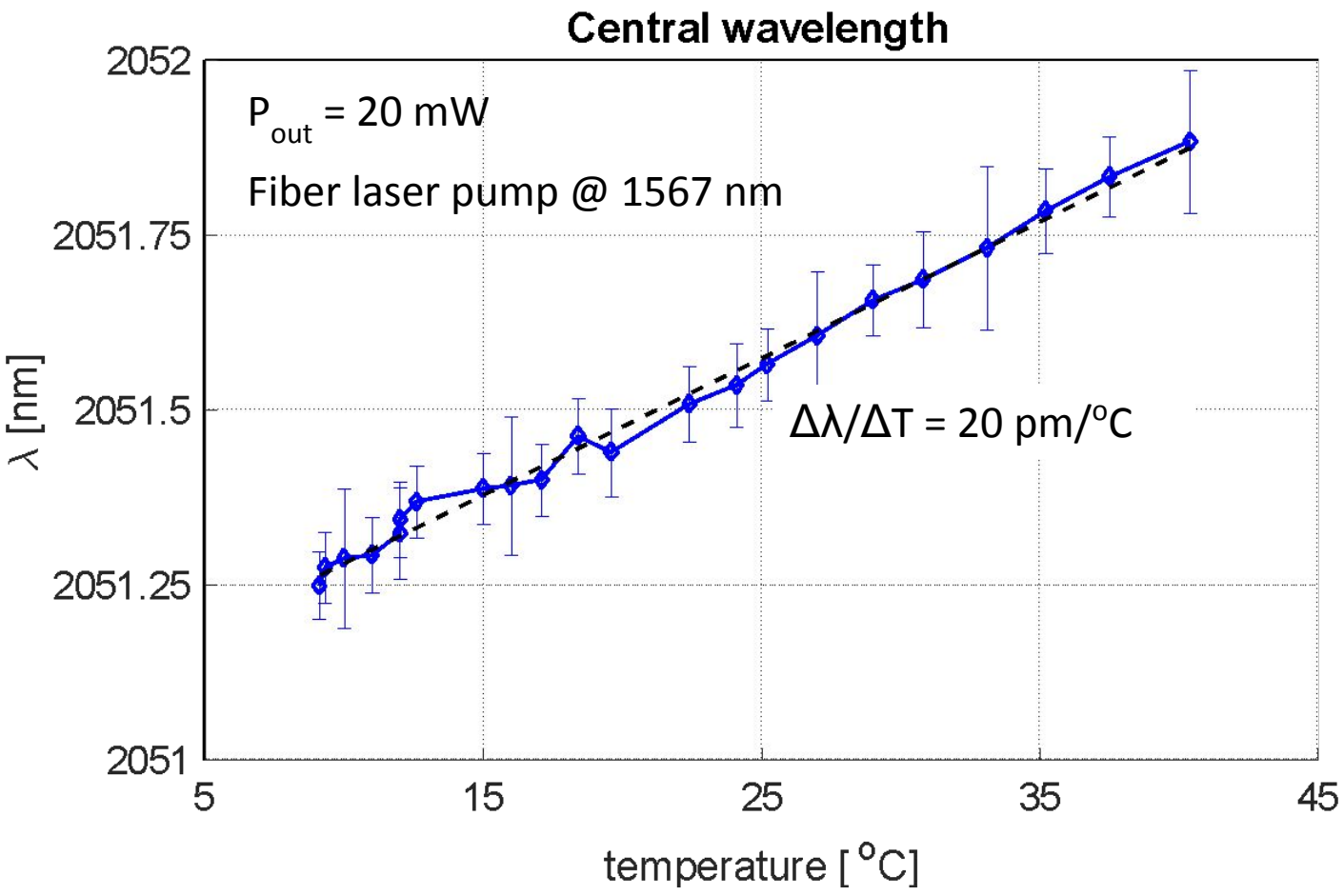




Thermal package enables:

- **Small wavelength shift 0.1 pm/mW of pump power**
- **Wavelength stability and single mode performance**

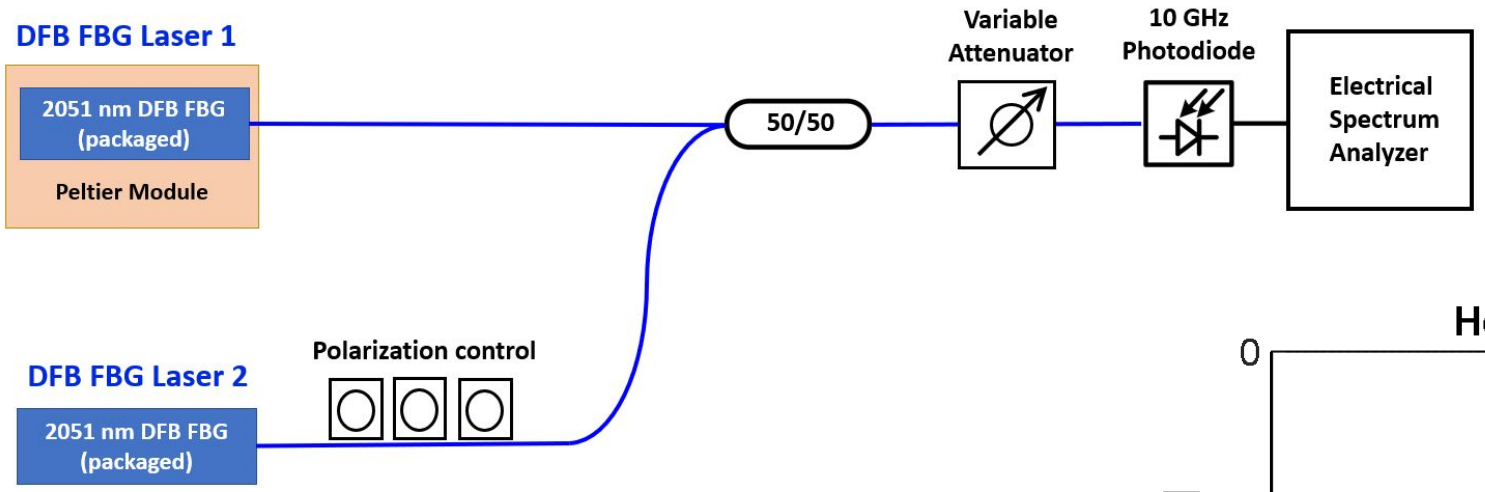
DFB-FBG wavelength dependence on temperature



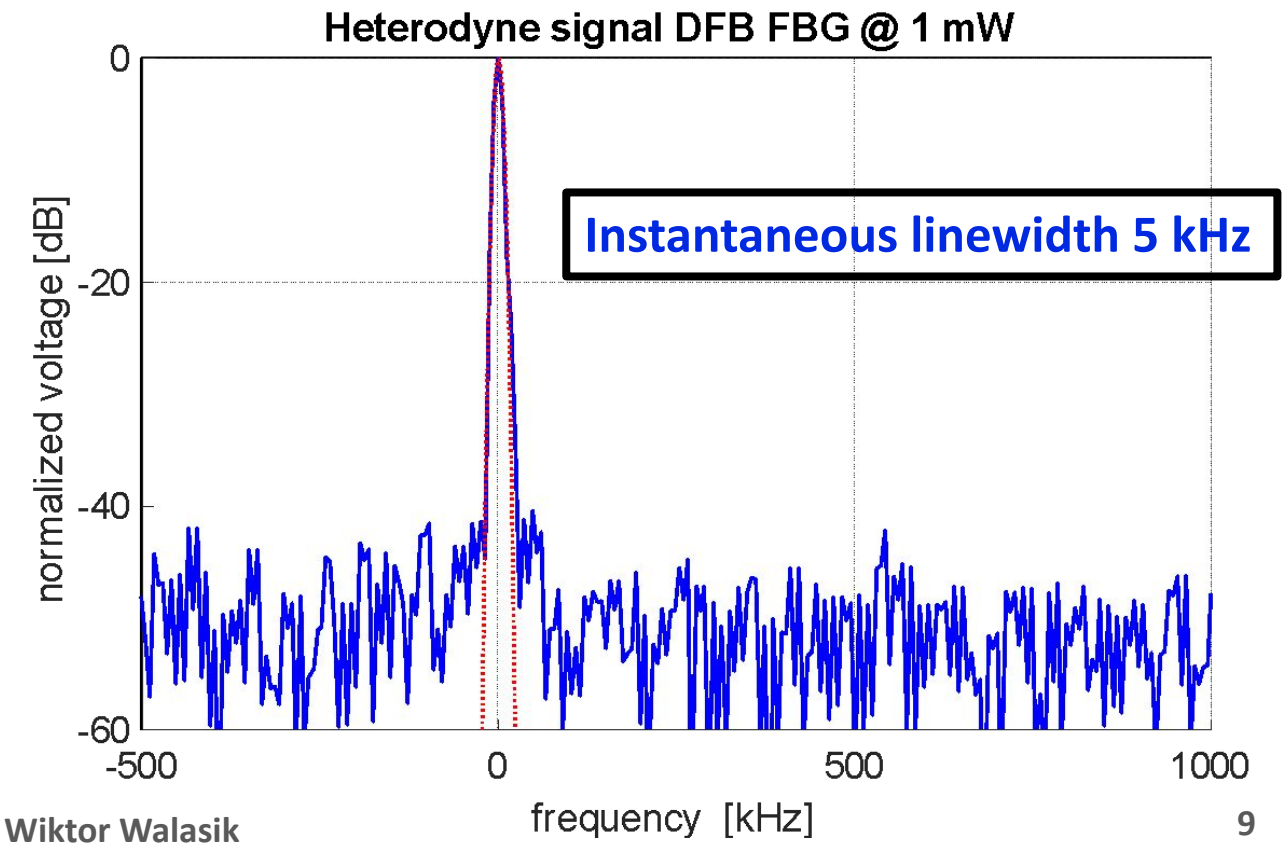
	DFB FBG	Semiconductor
Temp. tuning [pm/°C]	20	100
[GHz/°C]	1.4	7.1

- **Better wavelengths stability compared to semiconductor DFB lasers**
- **Tunability by 0.6 nm**

Heterodyne linewidth measurement

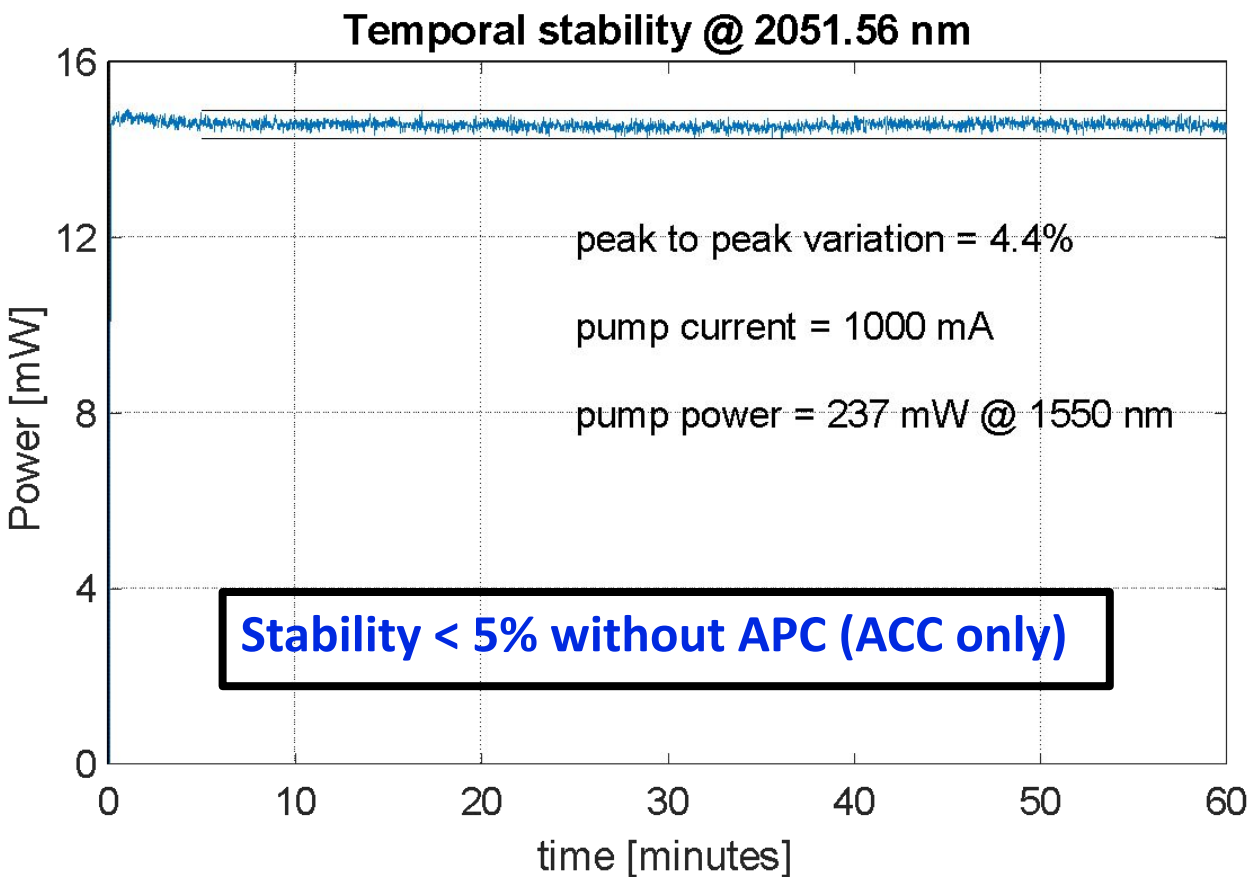


Heterodyne setup



5-kHz linewidth independent of wavelength and pump power

DFB-FBG laser power stability

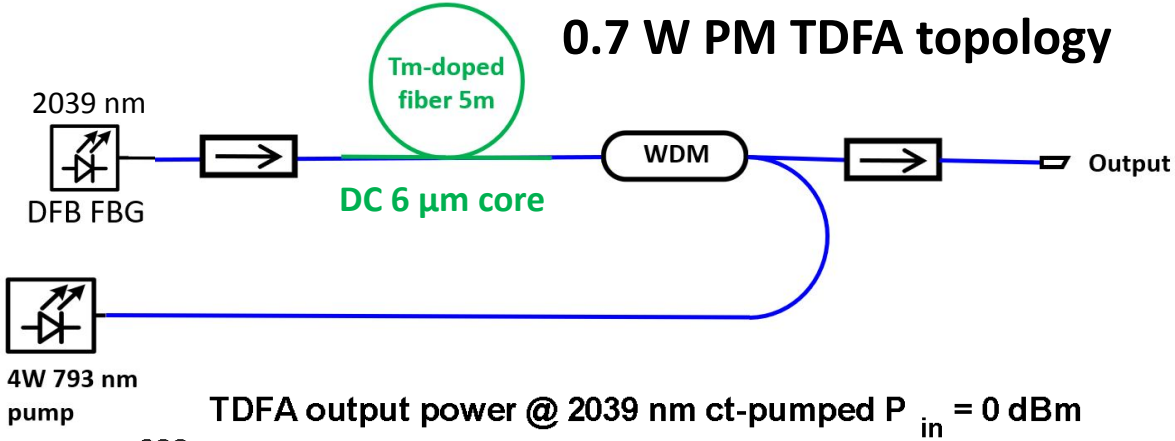


Future measurements:

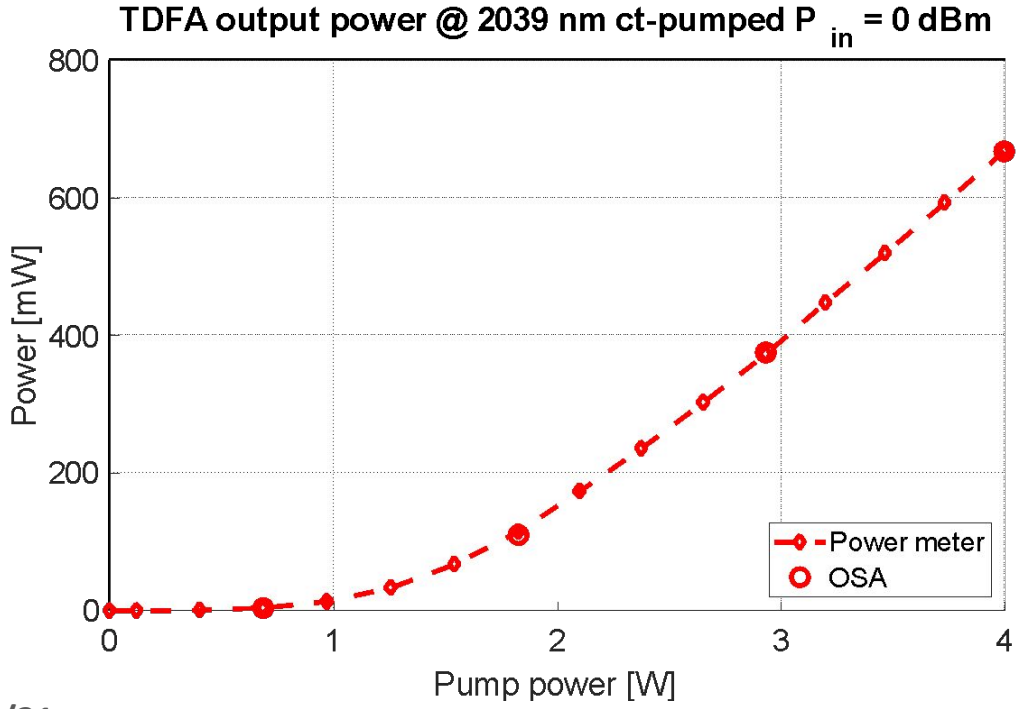
- **1/f noise (with SC and FL pumps)**
- **Short term stability**

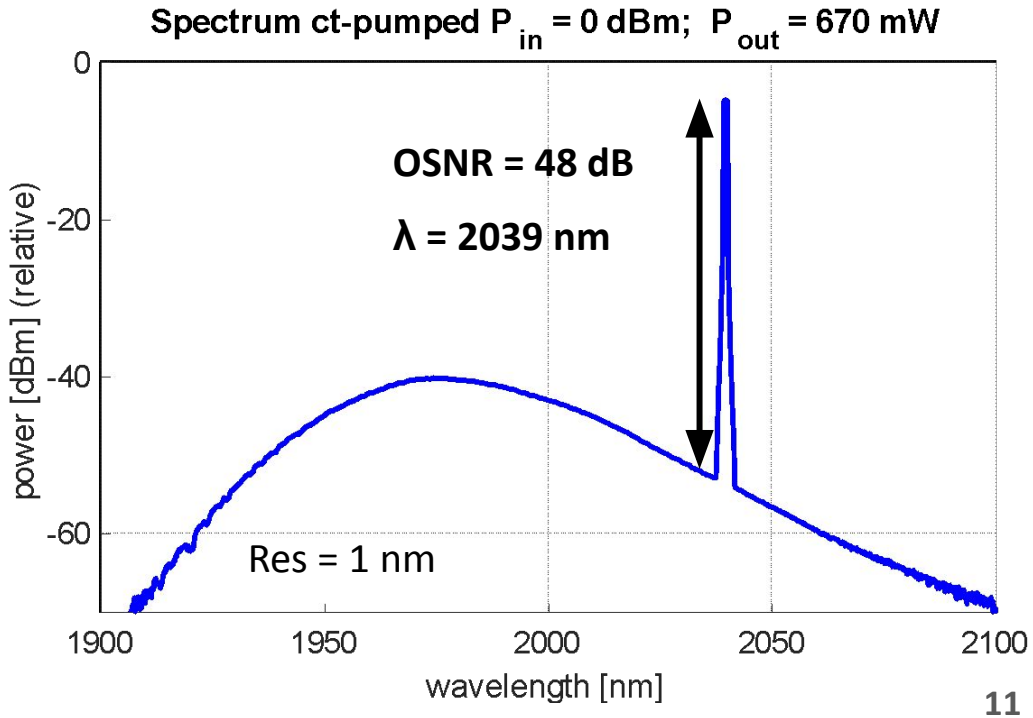
Amplification of DFB-FBG signal with a PM TDFA

0.7 W PM TDFA topology

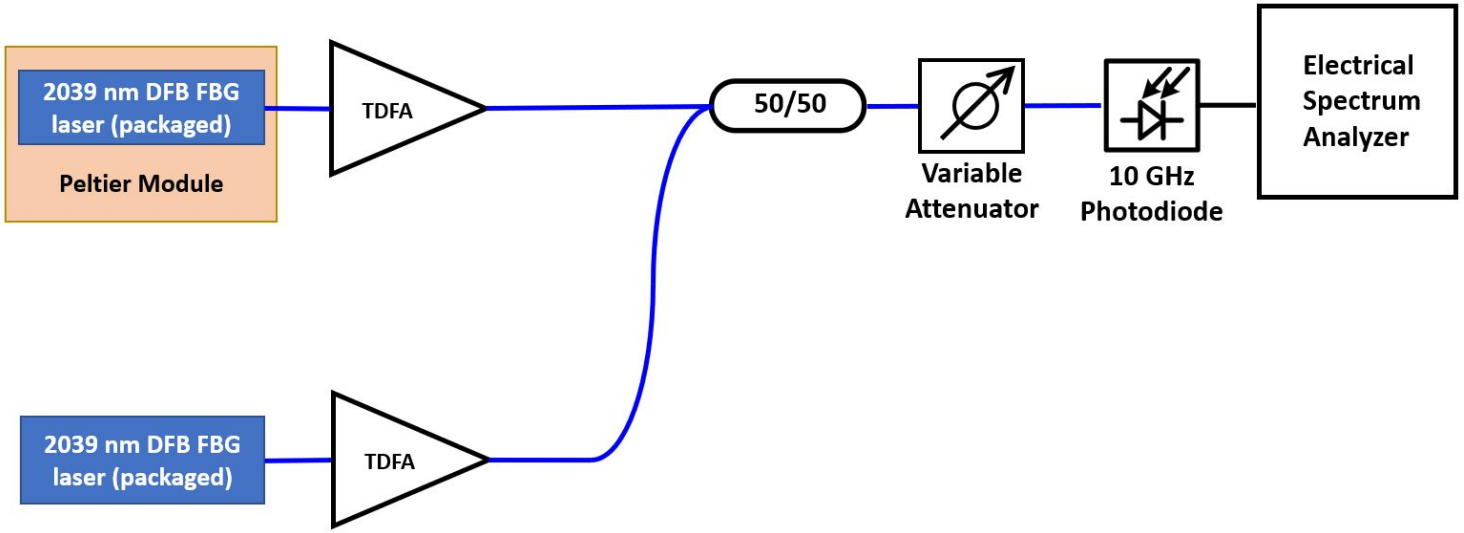


- Up to 670 mW of stable output power
- OSNR > 48 dB

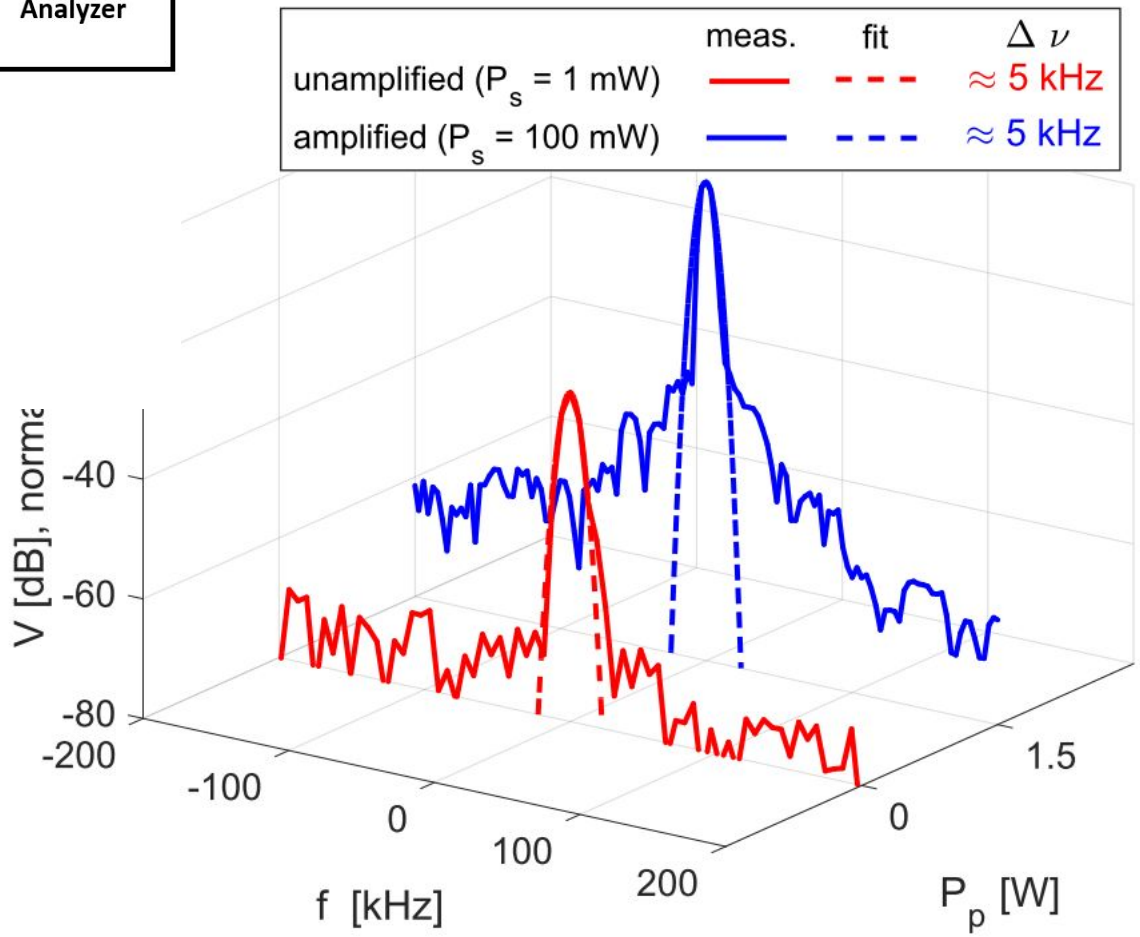




Amplified linewidth

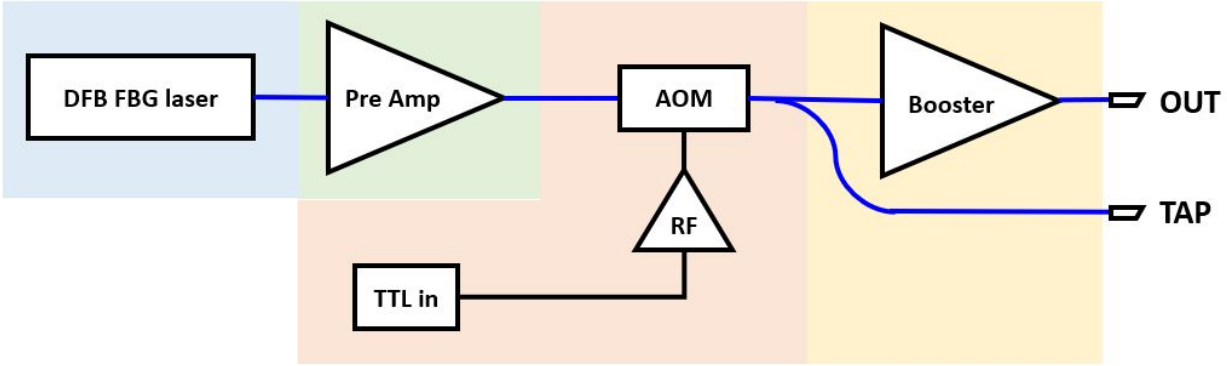


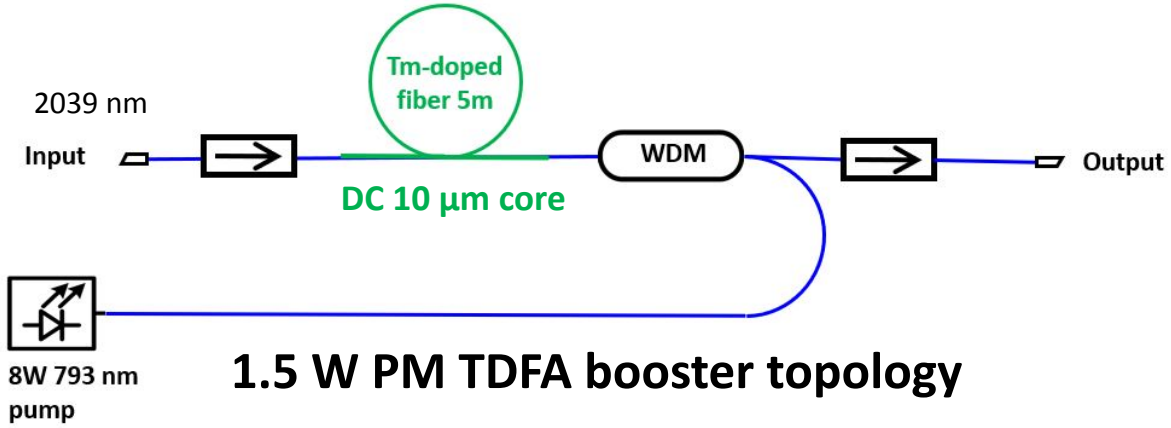
Heterodyne setup



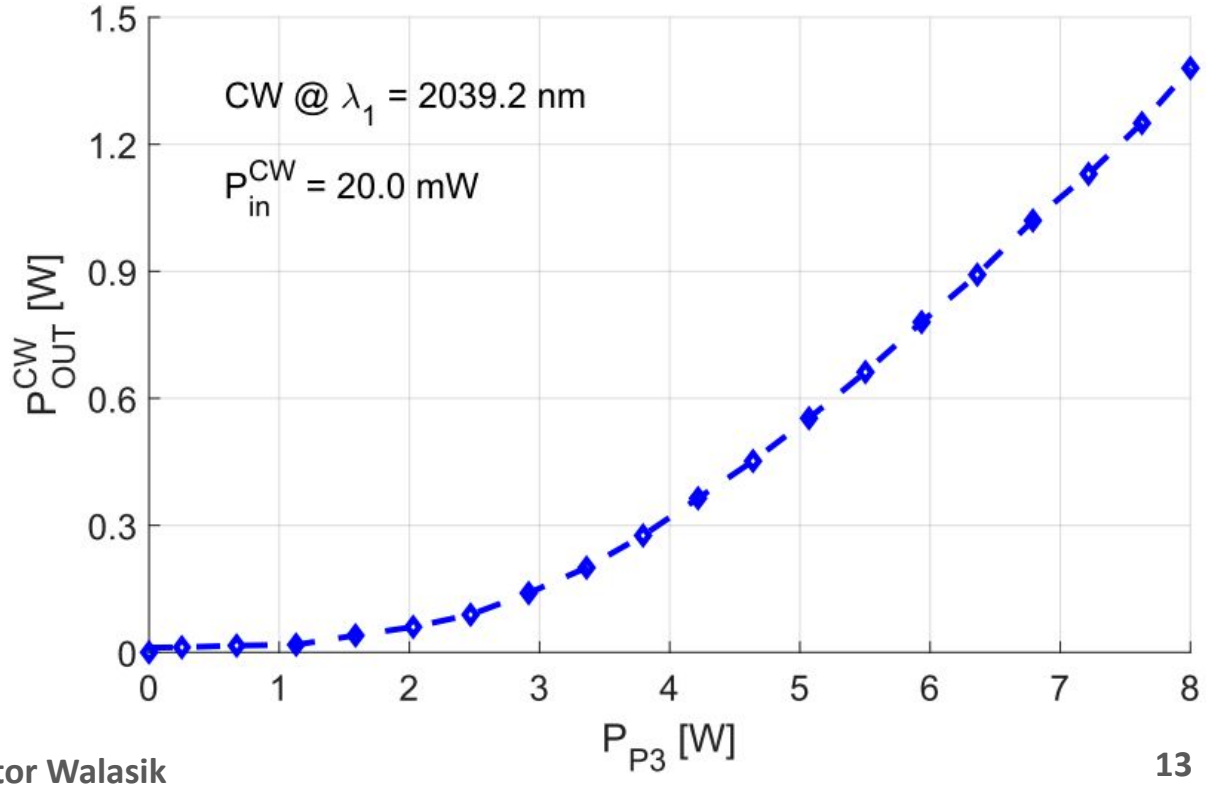
- **Instantaneous linewidth 5 kHz**
- **NO linewidth degradation upon amplification!**

Amplification of DFB-FBG signal with a dual stage PM TDFA



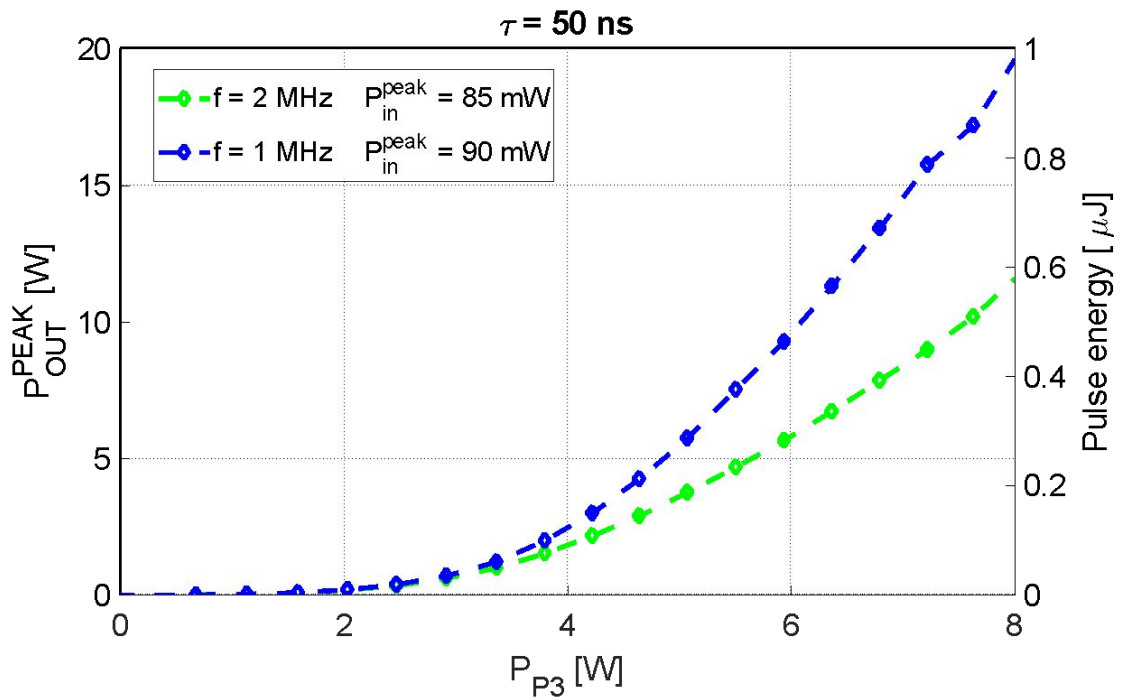


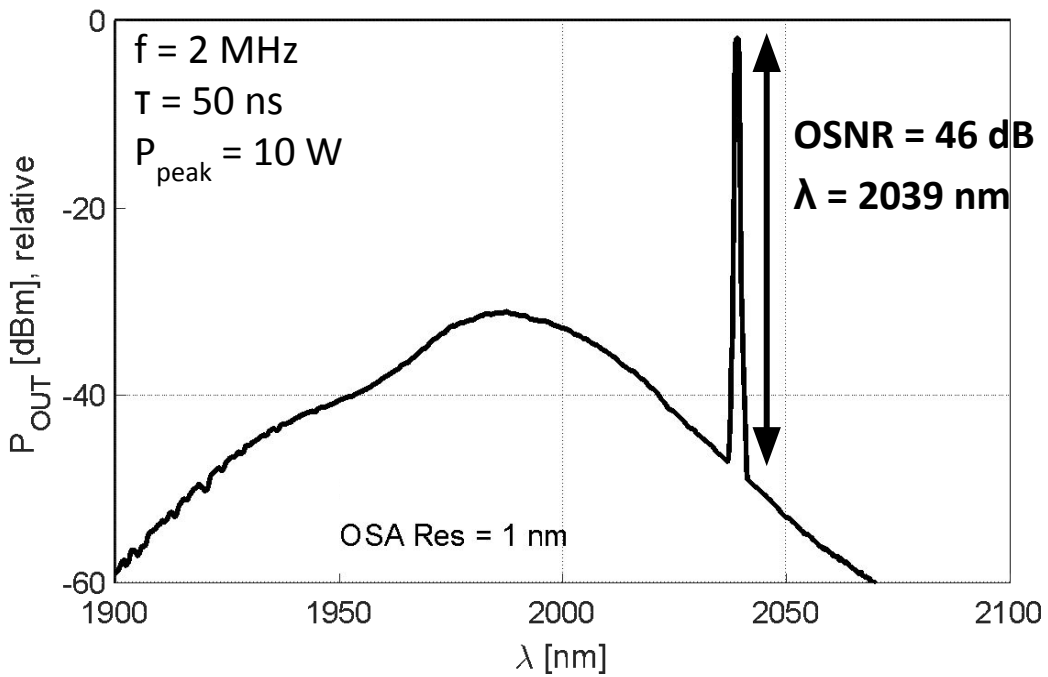
1.5 W PM TDFA booster topology



- More than 1.4 W CW of stable output power
- No SBS observed

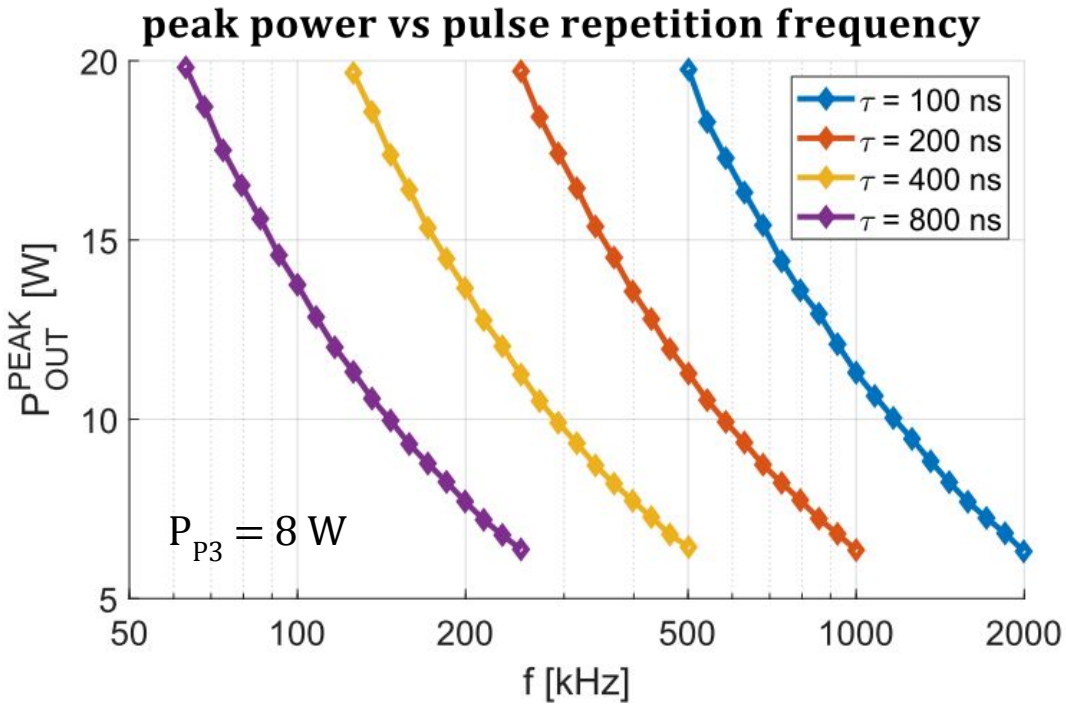
50 ns 1-2 MHz pulsed performance

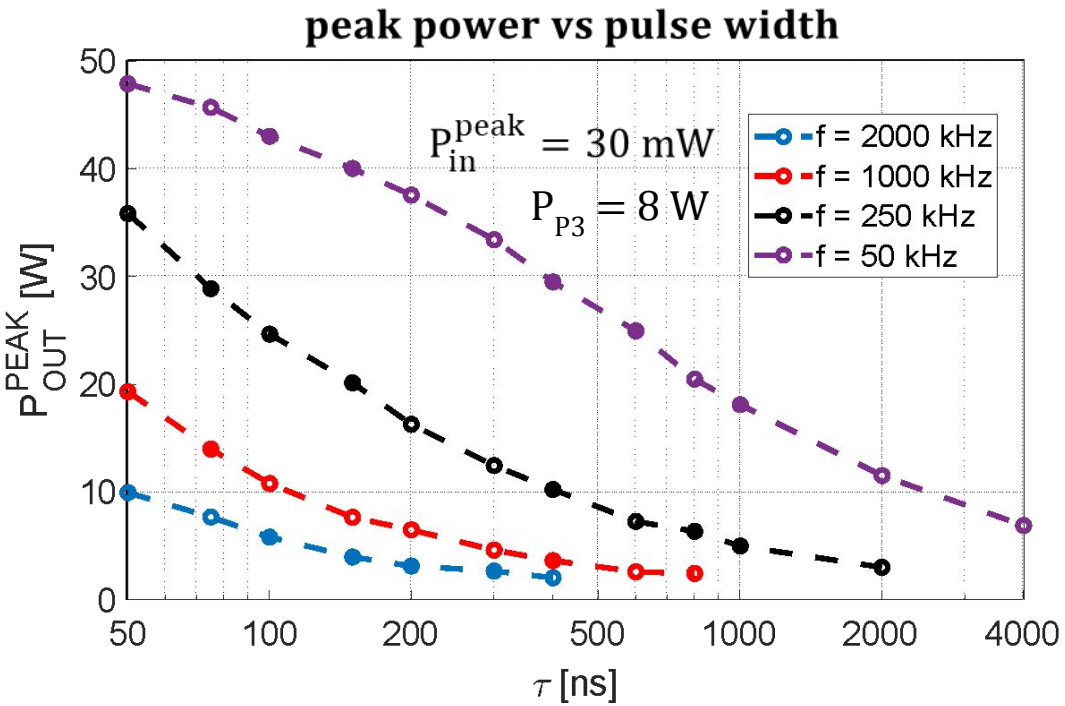




- **20 W peak power for 50 ns pulses at PRF of 1 MHz -> 1µJ pulse energy**
- **OSNR > 45 dB/1nm @ 5% duty cycle**
- **No nonlinear effects observed**

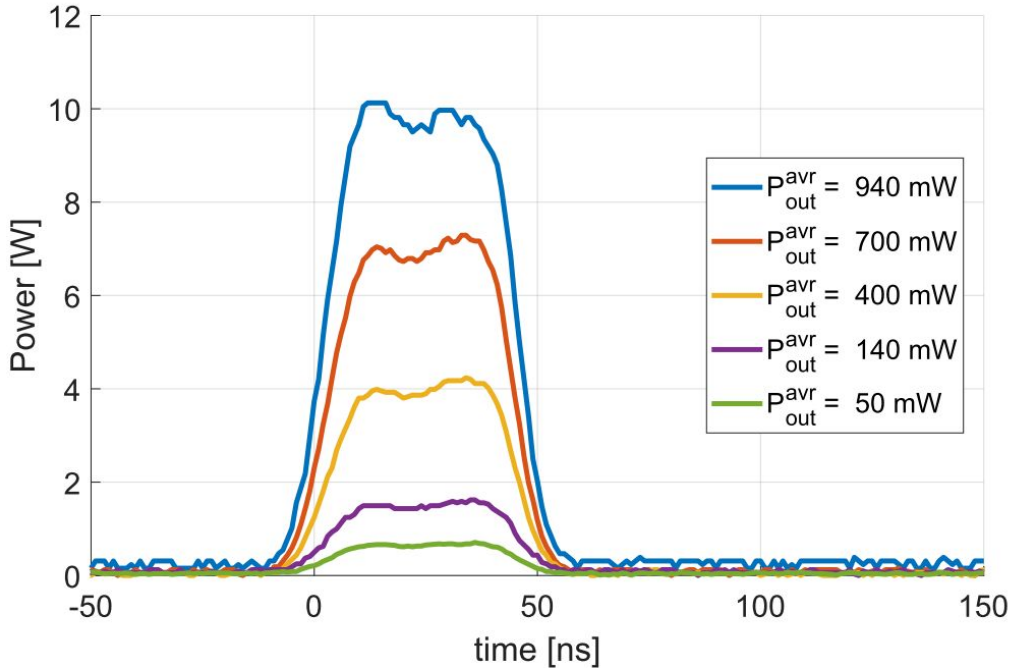
Power Booster: Parametric study

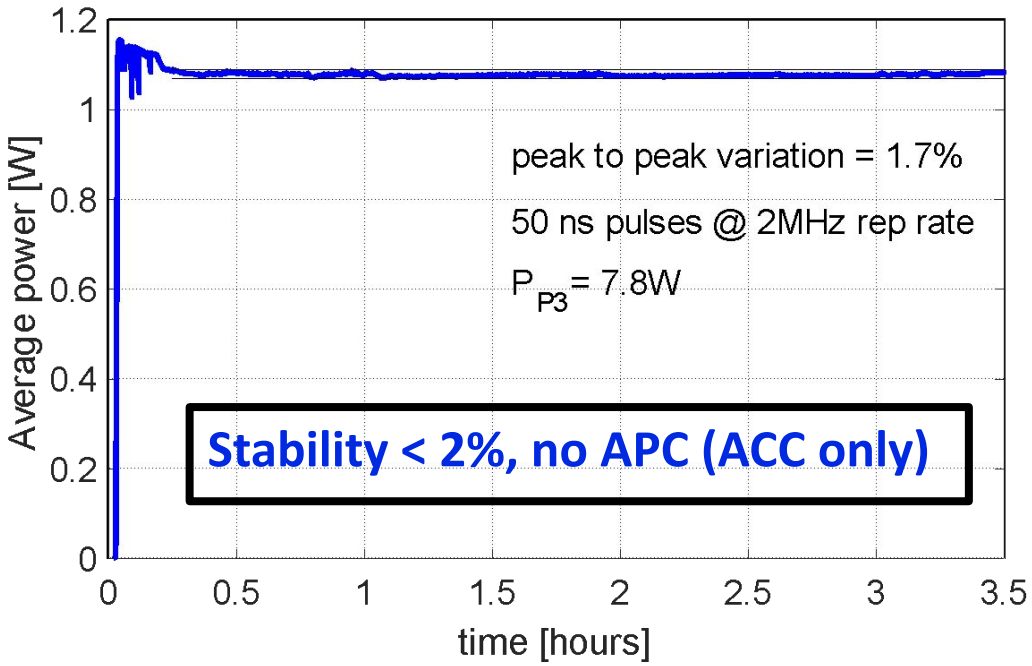




- **Versatile performance: P_{avr} > 1 W over wide range of PRFs and pulse durations**
- **Pulse energy > 27 µJ for 4 µs pulses at PRF of 50 kHz**
- **P_{peak} > 45 W @ 50 kHz PRF**

Pulse shape and stability





- **No pulse shape distortion**
- **No SBS observed for PRF > 50 kHz**

Conclusions

19’’ 2U benchtop

5-kHz PM DFB-FBG as a seed source:

- High output power: up to 60 mW
- Stable wavelength with low thermal drift
- Linewidth of 5 kHz before and after amplification
- CW Amplification to 1.4 W with a PM TDFA without distortion
- 50 W peak power and 27 µJ of pulse energy
- No nonlinear fiber interactions observed (i.e., Stimulated Brillouin Scattering) over broad range of pulse lengths (50 ns – 4µs) and PRF (50 kHz – 2 MHz)
- Promising source for 2 µm applications

Thank you for your attention